DAEGU APARTMENT PRICE PREDICTION AND OPTIMIZATION

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Outlines

01

Business Understanding

Presents the background context, business problem, objectives, and the analytical approach with evaluation metrics used to develop and assess a machine learning model for predicting apartment prices.

02

Data Understanding and Preparation

Details the process of understanding the dataset, followed by data cleaning, exploratory analysis, and preprocessing to ensure the data is accurate, consistent, and ready for modeling.

03

Modeling and Evaluation

Focuses on developing and optimizing benchmark models, selecting the best-performing model, and conducting diagnostics and interpretation to ensure reliable and insightful results.

04

Conclusion and Recommendations

The conclusion summarizes key findings and project limitations, while the recommendations focus on pricing strategies and model improvements.

BUSINESS UNDERSTANDING

Presents the background context, business problem, objectives, and the analytical approach with evaluation metrics used to develop and assess a machine learning model for predicting apartment prices.

Background

Daegu's Housing Landscape

- Daegu, **the fourth-largest metropolitan city** in South Korea, has **over 2.1 million residents** as of 2025.
- Due to urbanization, there's **rising demand** for **high-rise apartments**, which now make up **57%** of all housing types.
- This growth **exerts pressure** on residential land and creates **complexity in the housing market**.

Price Volatility

- In 2024**, apartment prices fell 4.08%** (Jan–Oct) due to **oversupply**, **high interest rates**, and **fluctuating demand**.
- Although some recovery occurred by year's end, **uncertainty remains**.
- **Price inconsistencies persist**, luxury listings claim up to 10M KRW/m², but **real data** shows an **average of 726,200 KRW/m²**.

Difficult Decisions

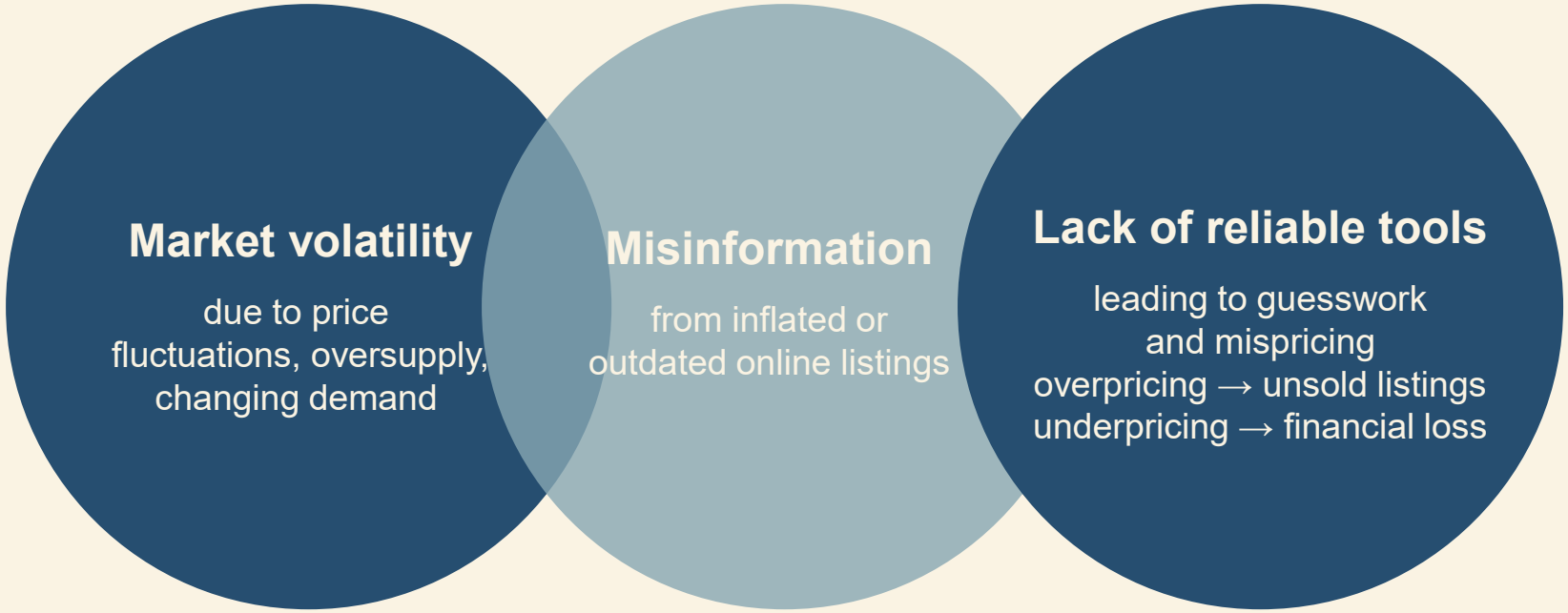
- Over 50% of apartments are 20+ years old, often needing renovation or lacking modern features.
- Meanwhile, limited new supply drives up prices in prime neighborhoods.
- **Owners face difficult decisions**, with many relying on **guesswork**, **outdated comparisons**, or **inflated online listings**.

Why We Need ML Model

- The problems lead to **frequent overpricing (unsold units)** or **underpricing (lost profits).**
- There's a clear **need** for a **data-driven pricing tool**.
- A **machine learning model** can provide **accurate, real-time price predictions**, empowering owners to make **confident** and **informed decisions**.

Business Problem

*Daegu **apartment owners** face challenges in setting the right apartment price due to:*



Objectives

PROJECT OBJECTIVES

To support **Daegu apartment owners** with a **machine learning–based pricing tool** that offers **reliable, data-informed estimates**, helping them set more appropriate prices **before listing** their properties.

01 **Provide accurate pricing guidance**
Use real transaction data to suggest fair, market-aligned prices.

02 **Reduce Pricing Risks**
Prevent overpricing or underpricing with model-backed reference ranges.

03 **Enhance Sales Efficiency**
Help attract buyers and speed up sales through optimal pricing.

04 **Promote Transparency**
Anchor prices to actual data for a clearer, more trustworthy market.

05 **Enable Scalable Predictions**
Design a model that can handle growing datasets and regular market updates.

ML OBJECTIVES

- Develop a reliable regression model
- Predict apartment prices accurately
- Analyze key features influencing price
- Ensure model interpretability

Analytic Approach and Metrics

ANALYTIC APPROACH

- This project develops a **regression model** using key apartment and location features to support competitive price estimation.
- The process includes data preparation, model benchmarking, tuning, and evaluation.

EVALUATION METRICS

Model performance is primarily assessed using **MAE** and **MAPE** for their interpretability and robustness, with **RMSE** included to capture larger errors.

DATA UNDERSTANDING AND PREPARATION

Details the process of understanding the dataset, followed by data cleaning, exploratory analysis, and preprocessing to ensure the data is accurate, consistent, and ready for modeling.

Data Understanding

The dataset contains **4,123 rows** and **11 columns**

Each row represents **a single apartment unit** and its associated features.

There are **10 feature** variables and **`SalePrice`** is the **target** variable

Feature	Description
HallwayType	Apartment hallway layout (corridor, terraced, mixed)
TimeToSubway	Walking time to nearest subway station
SubwayStation	Name of the nearest subway station
Facilities (ETC)	Number of nearby general facilities (stores, parks, etc.)
PublicOfficeNearby	Number of nearby public offices or government buildings
UnivNearby	Number of universities nearby
BasementParking	Number of basement parking spaces in the complex
YearBuilt	Construction year of the apartment
InAptFacilities	Number of on-site facilities (elevators, gyms, pools, etc.)
Size (sqft)	Total floor area of the apartment (in square feet)
SalePrice	Apartment selling price (in Korean Won) – *Target variable*

Feature variables (X)

Target variable (y)

Data Cleaning

Incorrect Data Types

- **Converted from float to integer format** for proper handling as count variables: N_FacilitiesNearBy(ETC), N_FacilitiesNearBy(PublicOffice), N_SchoolNearBy(University), N_Parkinglot(Basement)
- Ensures correct interpretation and improves regression performance.

Missing Values

- **No missing values detected** in the dataset.
- No imputation or removal needed.
- Ensures data completeness and reliability.

Duplicate Data

- Found and removed **1,422 duplicate rows**.
- Dataset reduced from **4,123** to **2,701** rows.
- Prevents data leakage, bias, and overfitting.

Inconsistent Data

- In **TimeToSubway** column:
  - **Corrected 1,248 format issues:**
    '0-5min' → '0~5min'
  - **Fixed 100 mislabeled categories:**
    'no_bus_stop_nearby' → 'no_subway_station_nearby'
- Standardized for semantic clarity and better encoding.

Outliers

- **Outliers found in**:
  - Size(sqft): **84 units**
  - SalePrice: **17 units**
- **Kept the outliers** due to relevance for large or luxury units, helping the model generalize across price ranges.

Feature Names

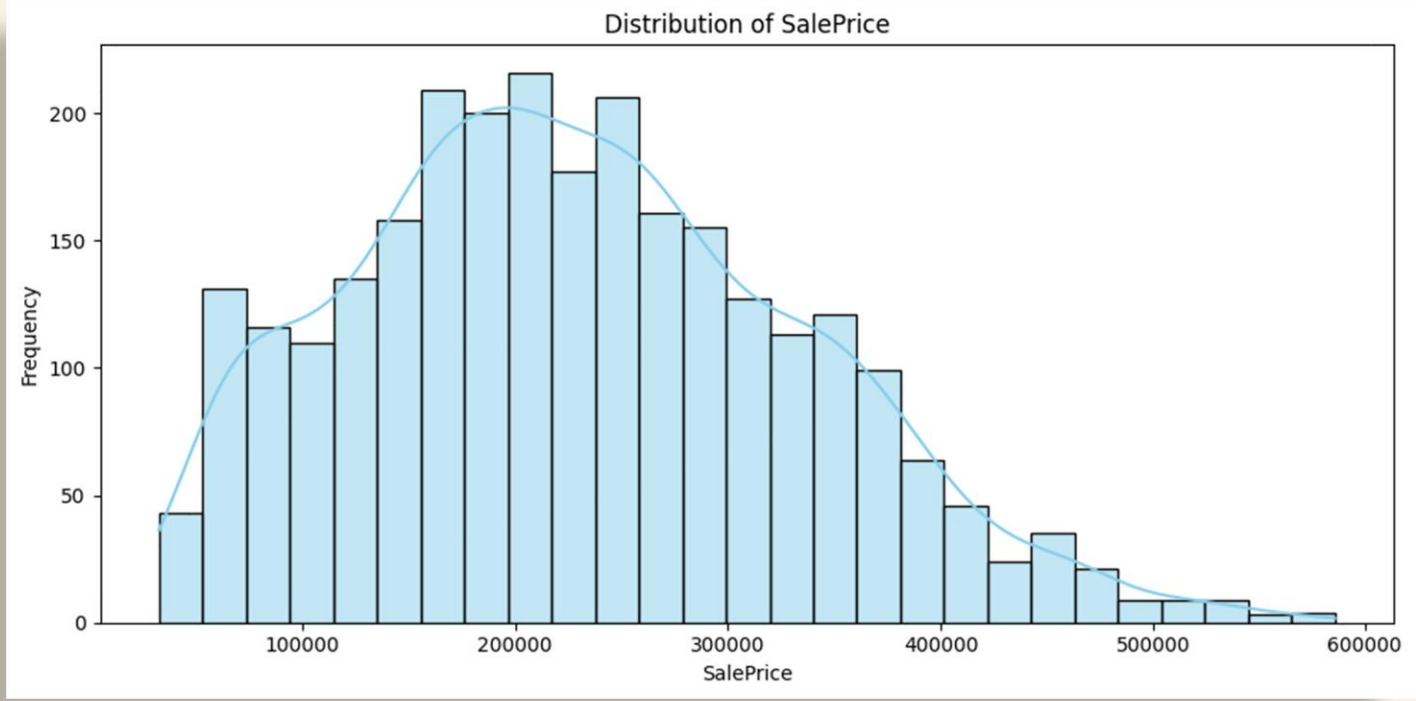
Renamed some feature names for clarity and consistency:

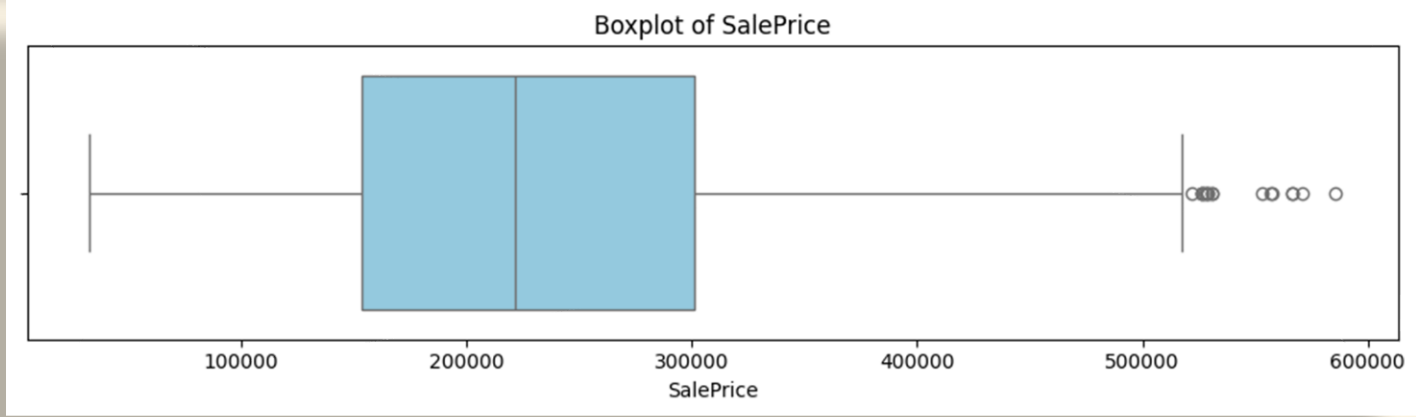
- N_FacilitiesNearBy(ETC) → NearbyGeneralFacilities
- N_FacilitiesNearBy(PublicOffice) → NearbyPublicOffices
- N_SchoolNearBy(University) → NearbyUniversities
- N_Parkinglot(Basement) → BasementParkingSpaces
- N_FacilitiesInApt → InApartmentFacilities
- Size(sqft) → SizeSqft

CLEANED DATASET OVERVIEW

Final Rows: 2,701
Final Columns: 11
Data is clean, but **further preprocessing** (e.g., scaling, encoding) **is still needed** for modeling.

Exploratory Data Analysis (EDA)

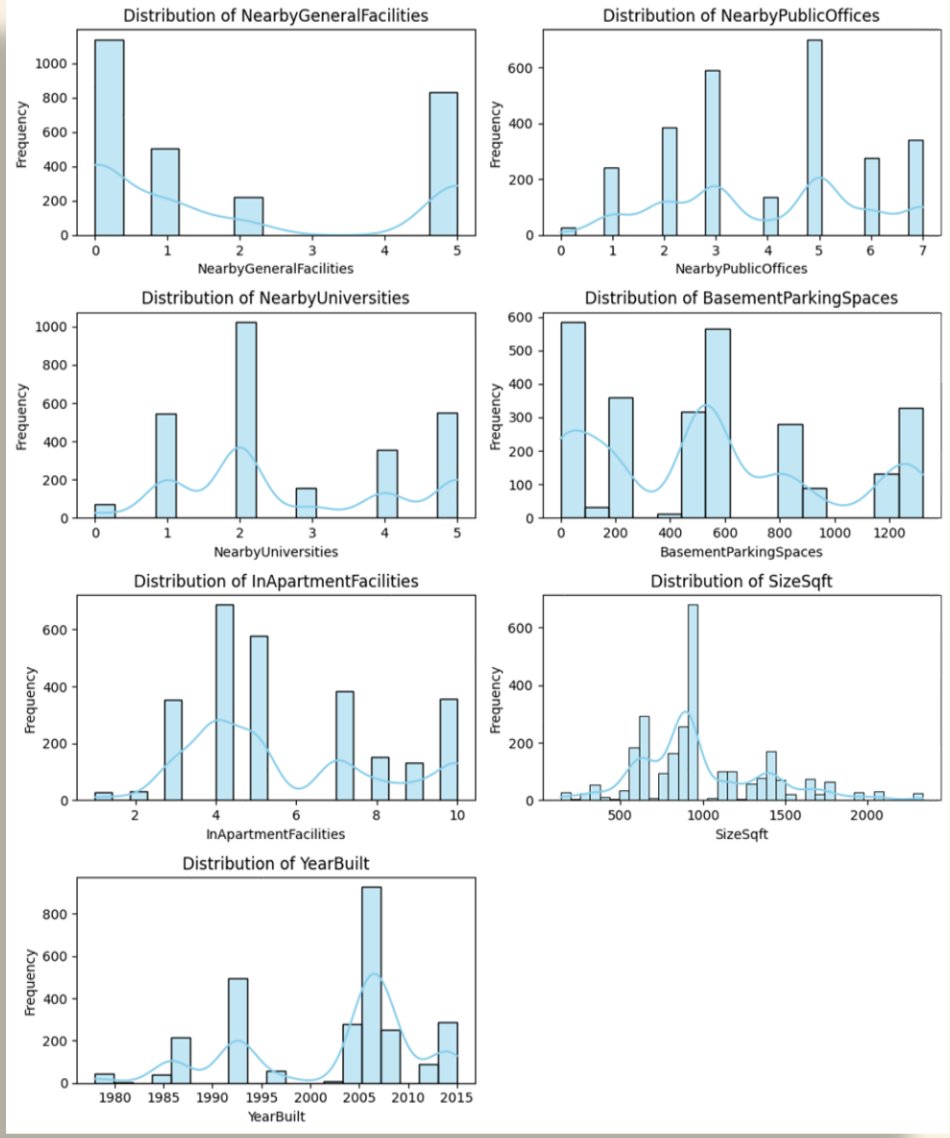


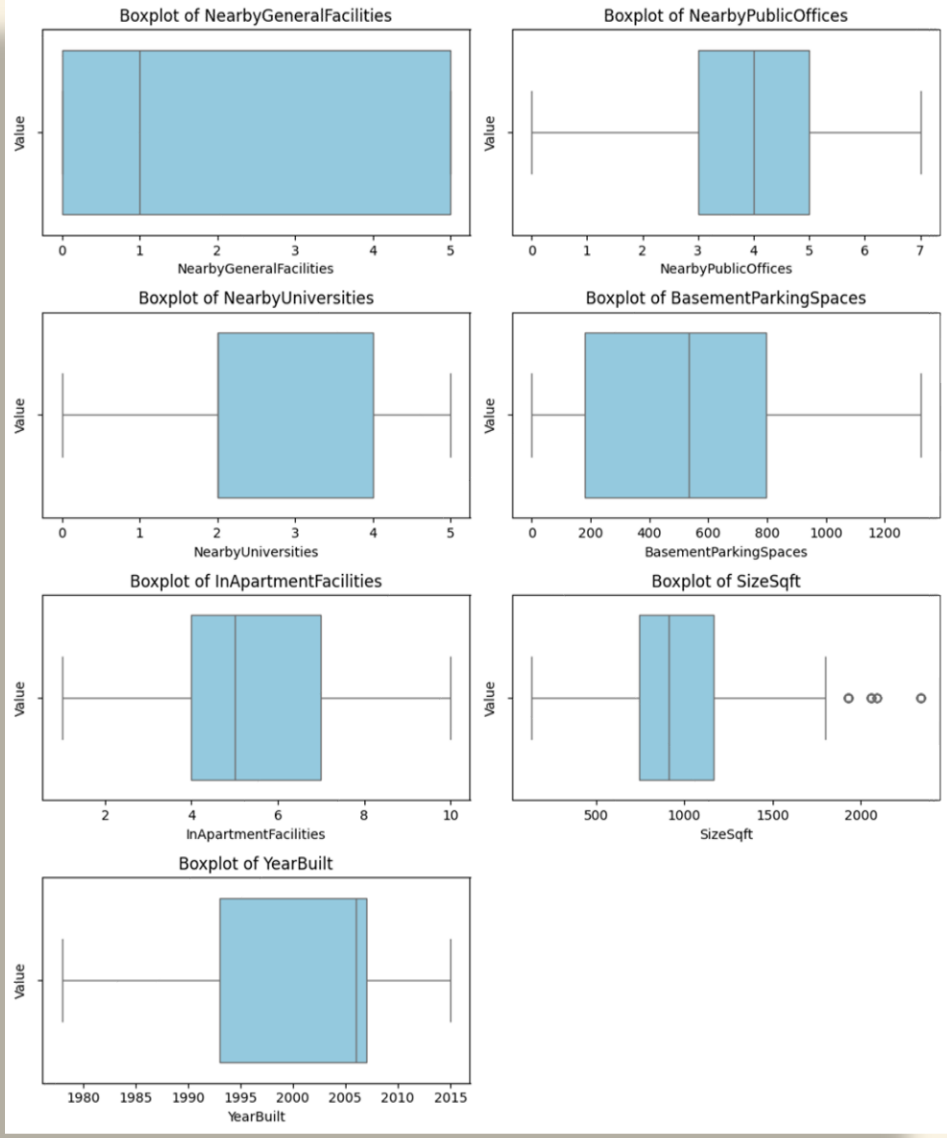


DISTRIBUTION OF TARGET VARIABLE

- **Right-Skewed & Imbalanced:** Most prices fall between ₩150K – ₩300K, skewing predictions toward the mid-range.
- **Outliers Exist:** High-end prices can distort predictions but may carry valuable patterns.
- **Wide Range & Uneven Errors:** around ₩30K–₩600K spread with increasing error at higher prices (heteroscedasticity).

Exploratory Data Analysis (EDA)

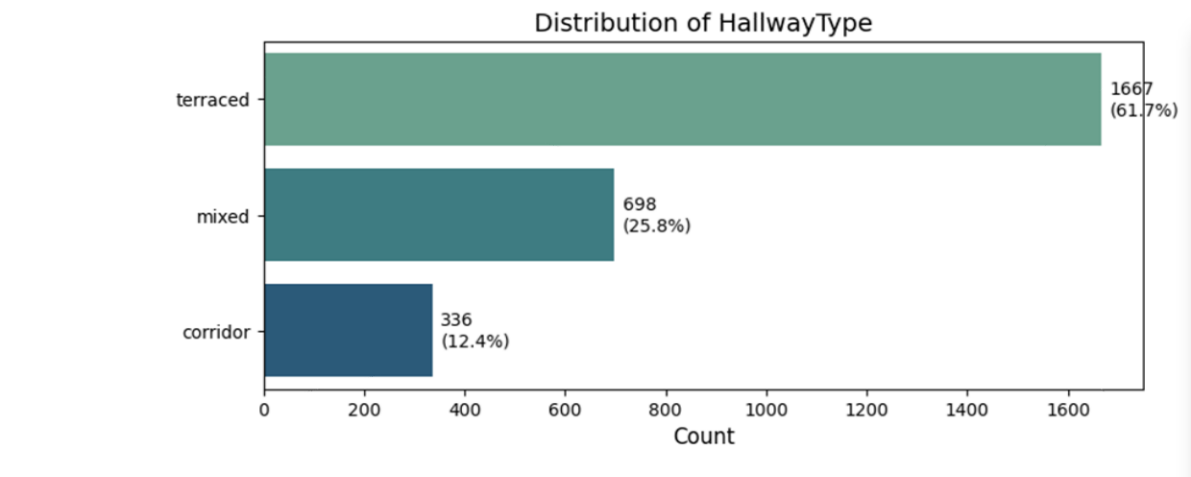


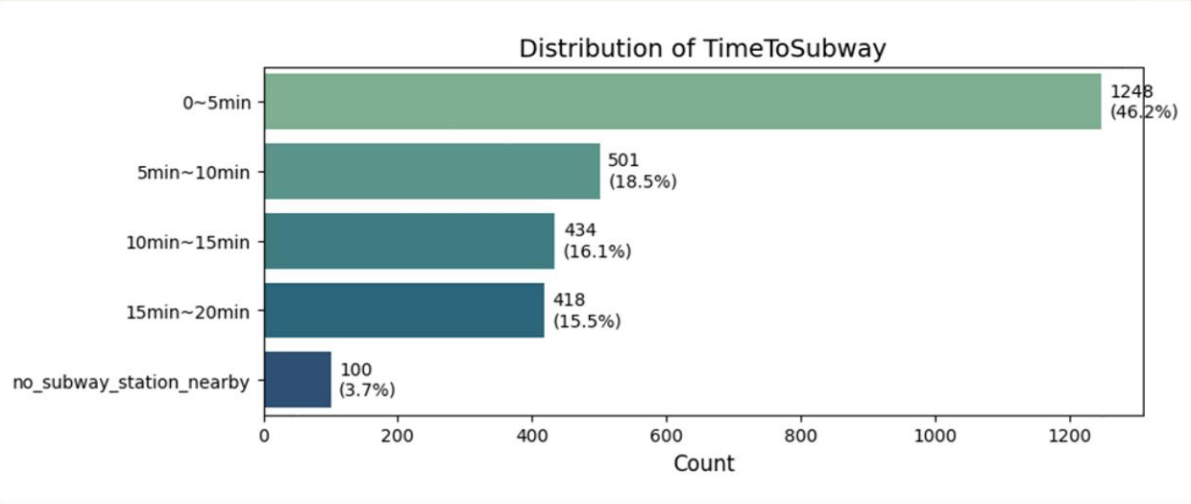


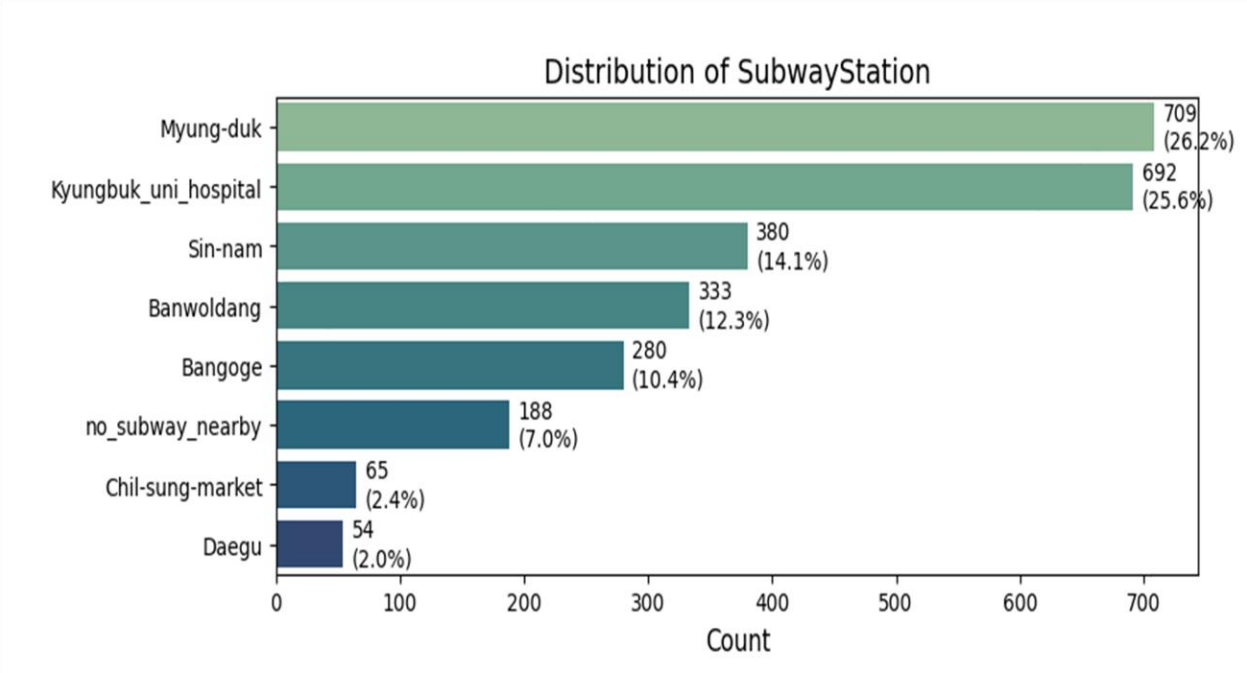
DISTRIBUTION OF NUMERICAL FEATURES

- **Skewed Distributions:** Features like ‘SizeSqft’ and ‘BasementParkingSpaces’ are right-skewed with extreme values.
- **Clustered Patterns:** Variables such as ‘InApartmentFacilities’ and ‘NearbyUniversities’ cluster around specific values, indicating usage patterns and gaps.
- **Outliers & Limited Spread:** ‘SizeSqft’ has outliers, while ‘YearBuilt’ and ‘NearbyGeneralFacilities’ show narrow, concentrated ranges.

Exploratory Data Analysis (EDA)



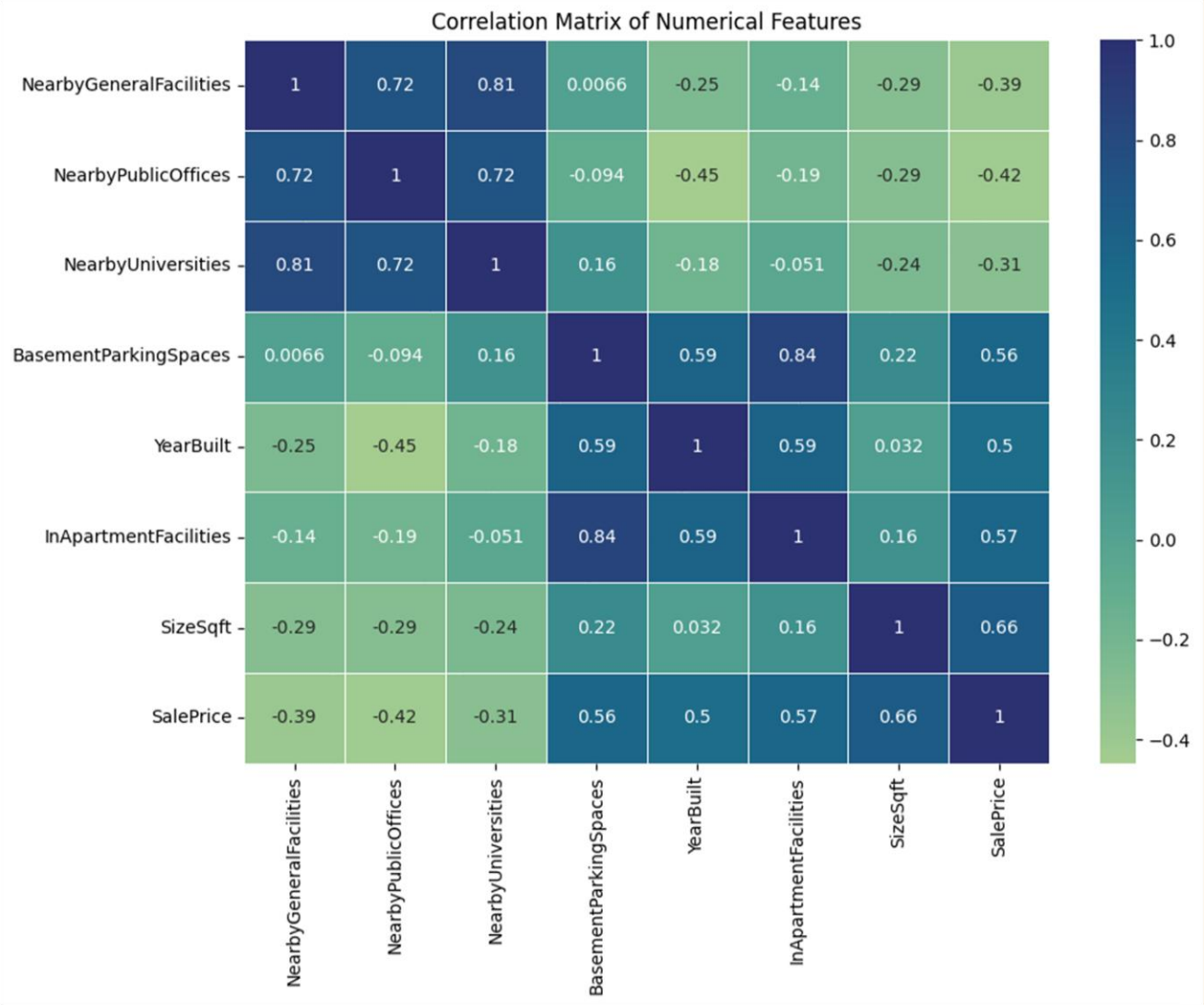




DISTRIBUTION OF CATEGORICAL FEATURES

- **Terraced Hallways:** 60%+ of apartments use this layout—Daegu's preferred design.
- **Subway Access:** 46%+ of listings are within 5 minutes of a station.
- **Station Clustering:** Most listings near Myung-duk & Kyungbuk Uni Hospital stations.

Exploratory Data Analysis (EDA)



FEATURE CORRELATION

- **Strong Price Predictors:** ‘SizeSqft’, ‘InApartmentFacilities’, ‘BasementParkingSpaces’, and ‘YearBuilt’ show the highest positive correlations with ‘SalePrice’ (0.50–0.66).
- **Urban Disamenities:** ‘NearbyPublicOffices’, ‘NearbyGeneralFacilities’, and ‘NearbyUniversities’ have moderate negative correlations (-0.31 to -0.42), possibly reflecting noise or congestion.
- **Feature Clustering:** Amenity features like ‘BasementParkingSpaces’ and ‘InApartmentFacilities’ are highly correlated (0.84+), revealing quality clusters that can influence model design.

Data Preprocessing

ROBUST SCALING — ONE-HOT ENCODING — ORDINAL ENCODING — BINNING

Feature	Transformation Type	Technique	Reason
NearbyGeneralFacilities	Scaling	Robust Scaler	Numeric, skewed distribution
NearbyPublicOffices	Scaling	Robust Scaler	Numeric, skewed distribution
NearbyUniversities	Scaling	Robust Scaler	Numeric, skewed distribution
BasementParkingSpaces	Scaling	Robust Scaler	Numeric, skewed distribution
InApartmentFacilities	Scaling	Robust Scaler	Numeric, skewed distribution
SizeSqft	Scaling & Binning	Robust Scaler & Binning (+Ordinal Encoder)	Numeric, skewed distribution with outliers; binned into 4 size-based categories
HallwayType	Encoding	One-Hot Encoder	Nominal categorical with fewer than 10 unique categories
SubwayStation	Encoding	One-Hot Encoder	Nominal categorical with fewer than 10 unique categories
TimeToSubway	Encoding	Ordinal Encoder	Ordinal categorical variable indicating proximity to subway
YearBuilt	Binning	Binning (+Ordinal Encoder)	Binned into 4 categories based on construction era

YearBuilt_Binned

Category	Interval (Years)	Description
Very Old	1977 < x ≤ 1987	Older buildings with potentially outdated standards
Old	1987 < x ≤ 1997	Transition phase with improved materials and design
Modern	1997 < x ≤ 2006	Buildings with more up-to-date styles and improved construction
New	2006 < x ≤ 2015	Recently built with modern designs and updated safety standards

SizeSqft_Binned

Category	Interval (Sqft)	Description
Small	134 < x ≤ 685	Compact apartment units
Medium	685 < x ≤ 1236	Moderately sized living spaces
Large	1236 < x ≤ 1787	Spacious apartments
Very Large	1787 < x ≤ 2337	High-end or premium-sized units

MODELING AND EVALUATION

Focuses on developing and optimizing benchmark models, selecting the best-performing model, and conducting diagnostics and interpretation to ensure reliable and insightful results.

Benchmark Model

Model	Description	MAE	MAPE	MSE	RMSE	RMSPE	RMSLE	R^2
Gradient Boosting	Builds trees sequentially, where each tree corrects the errors of the previous one.	35,891.85	0.184356	1.97E+09	44,432.71	0.232548	0.220352	0.813677
XGBoost	An optimized version of gradient boosting with regularization and speed improvements.	35,935.25	0.184933	1.98E+09	44,463.12	0.233643	0.220923	0.813422
Bagging	Trains multiple models on random subsets of data and averages their predictions.	35,977.23	0.185734	1.98E+09	44,510.66	0.237164	0.222336	0.813023
CatBoost	Gradient boosting model that handles categorical variables automatically and efficiently.	36,458.92	0.188896	2.01E+09	44,794.74	0.236818	0.223215	0.810629
Random Forest	Combines many decision trees trained on random features and samples to reduce overfitting.	37,685.40	0.199017	2.18E+09	46,662.57	0.265148	0.235582	0.794507
Decision Tree	Splits data into branches based on feature thresholds, forming a tree-like model.	38,424.87	0.192783	2.41E+09	49,053.52	0.276615	0.239437	0.772909
KNN	Predicts based on the average target value of the ‘k’ closest training samples.	38,643.95	0.199182	2.31E+09	48,081.60	0.256314	0.239198	0.781818
Multiple Linear Reg.	Models the target as a linear combination of multiple input features.	41,175.25	0.220742	2.61E+09	51,069.03	0.307057	0.257551	0.753864
Lasso Regression	Linear regression with L1 regularization that can shrink some coefficients to zero.	41,180.72	0.220792	2.61E+09	51,070.53	0.307097	0.257563	0.753849
Ridge Regression	Linear regression with L2 regularization that reduces model complexity by shrinking coefficients.	41,254.47	0.221421	2.61E+09	51,096.33	0.307646	0.257802	0.753601

80:20 DATA SPLIT

Hyperparameter Tuning

BENCHMARK VS. BEST HYPERPARAMETER

Model	Benchmark Hyperparameters	Best Hyperparameters
Multiple Linear Regression	— (no tunable hyperparameters)	— (no tunable hyperparameters)
Lasso Regression	alpha=1.0	alpha=0.01
Ridge Regression	alpha=1.0	alpha=0.01
KNN Regressor	n_neighbors=5, weights='uniform'	n_neighbors=19, weights='distance', metric='minkowski'
Decision Tree Regressor	criterion='absolute_error', max_depth=5	max_depth=10, min_samples_split=15, min_samples_leaf=2, criterion='squared_error'
Bagging Regressor	n_estimators=100	n_estimators=200, max_samples=1.0, bootstrap=True, oob_score=True
Random Forest Regressor	n_estimators=100, max_depth=5	n_estimators=100, max_depth=10, min_samples_split=15, min_samples_leaf=2, max_features='sqrt', bootstrap=True, oob_score=True
Gradient Boosting Regressor	n_estimators=100, learning_rate=0.1, max_depth=5	n_estimators=100, learning_rate=0.05, max_depth=5, min_samples_split=5, min_samples_leaf=2, max_features='log2', subsample=1.0
XGBoost Regressor	n_estimators=100, learning_rate=0.1, max_depth=5	n_estimators=200, learning_rate=0.1, max_depth=7, min_child_weight=1, gamma=0.3, colsample_bytree=0.8, subsample=1.0
CatBoost Regressor	iterations=100, learning_rate=0.1, depth=6	iterations=300, learning_rate=0.1, depth=4, random_strength=0.5, l2_leaf_reg=5, bagging_temperature=0

PERFORMANCE COMPARISON: BENCHMARK VS. TUNED

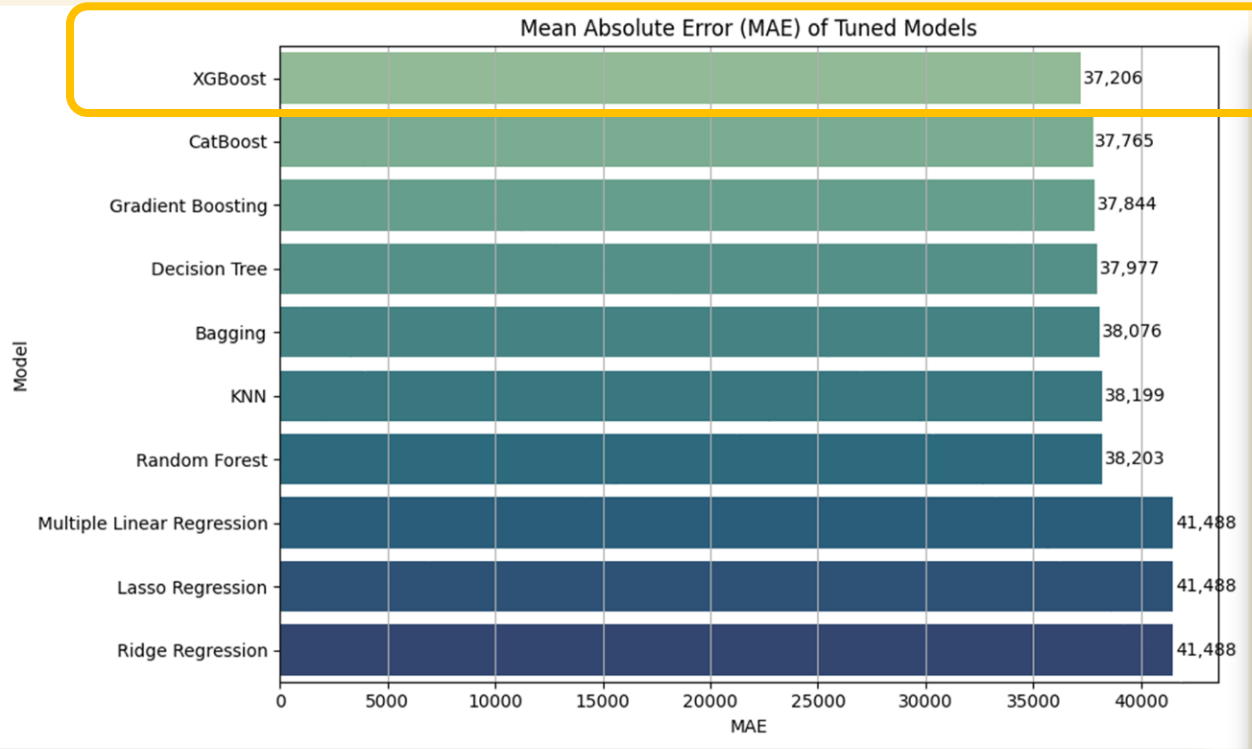
Model	MAE (B)	MAE (T)	MAPE (B)	MAPE (T)
GB	35,892	37,844	0.1844	0.1925
XGB	35,935	37,206	0.1849	0.1809
Bag	35,977	38,076	0.1857	0.1927
Cat	36,459	37,765	0.1889	0.1909
RF	37,685	38,203	0.199	0.1948
DT	38,425	37,977	0.1928	0.1918
KNN	38,644	38,199	0.1992	0.1915
MLR	41,175	41,488	0.2207	0.2131
Lasso	41,181	41,488	0.2208	0.2131
Ridge	41,254	41,488	0.2214	0.2131

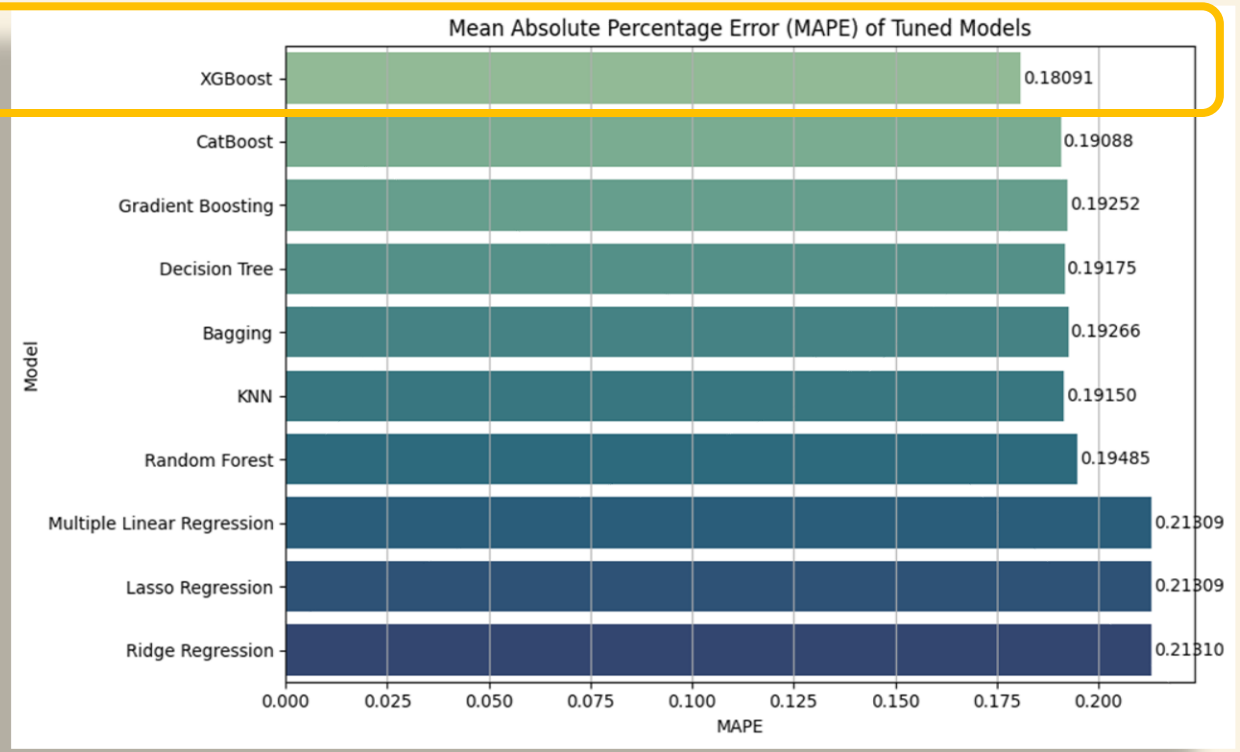
Notes:

- B = Benchmark, T = Tuned
- GB = Gradient Boosting, XGB = XGBoost, Bag = Bagging, Cat = CatBoost, RF = Random Forest, DT = Decision Tree, KNN = K-Nearest Neighbors, MLR = Multiple Linear Regression

Best Model Selection

MODEL PERFORMANCE AFTER HYPERPARAMETER TUNING

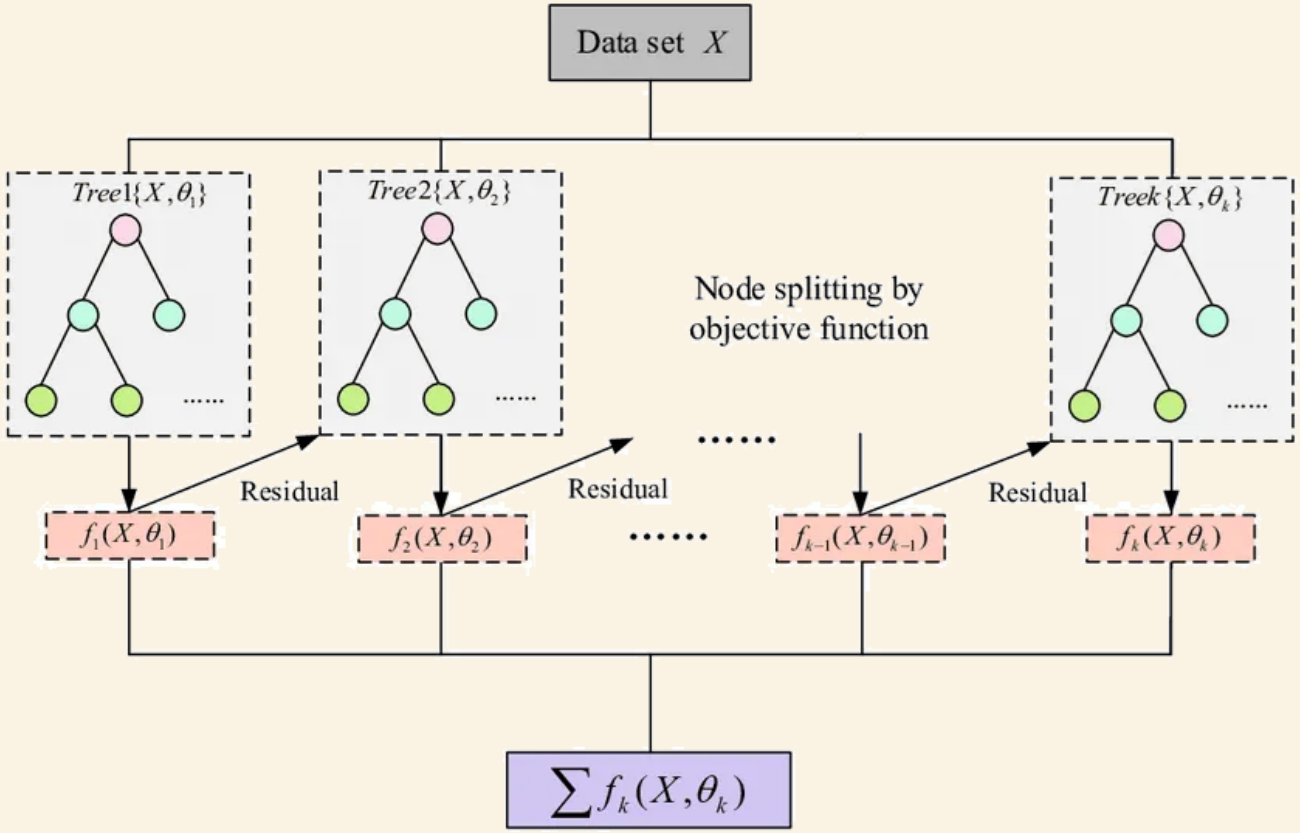




Final Model Selection:
XGBoost *was selected as the optimal model after extensive benchmarking and hyperparameter tuning. It combines* ***low error rates****,* ***stable cross-validation performance****, and practical strengths like* ***fast inference*** *and* ***model transparency****, making it highly reliable and scalable for apartment price prediction.*

Best Model: XGBoost

XGBoost is an **advanced gradient boosting** method that improves efficiency, scalability, and accuracy through **second-order optimization, regularization, pruning, parallelism,** and **effective handling of missing data**.



HOW IT WORKS

1. Sequential Tree Learning:

XGBoost builds trees one by one, each fixing errors from the previous model. A learning rate controls how much each tree impacts the final prediction for stability.

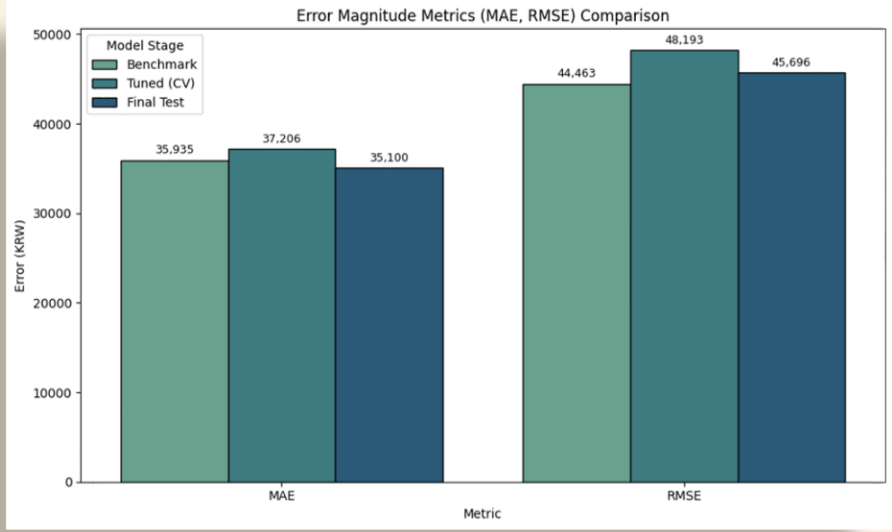
2. Regularization and Pruning:

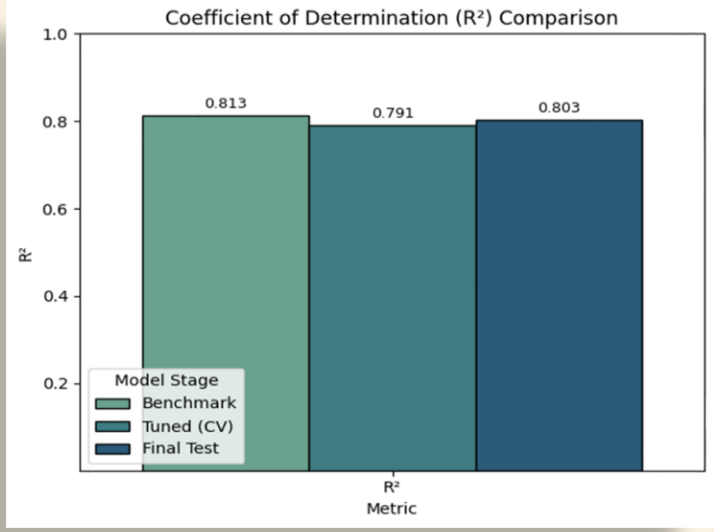
It prevents overfitting by shrinking leaf weights with L1/L2 regularization and pruning branches that don't improve the model, ensuring simpler, more generalizable trees.

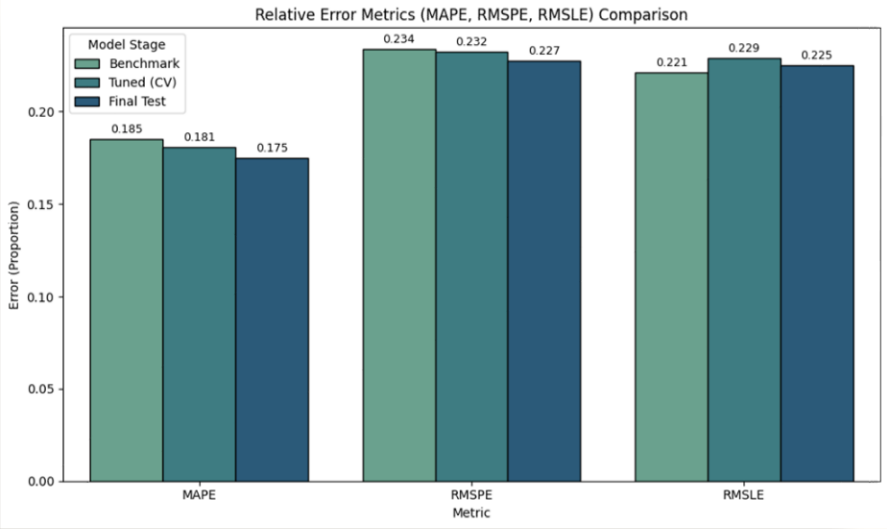
3. Fast and Scalable:

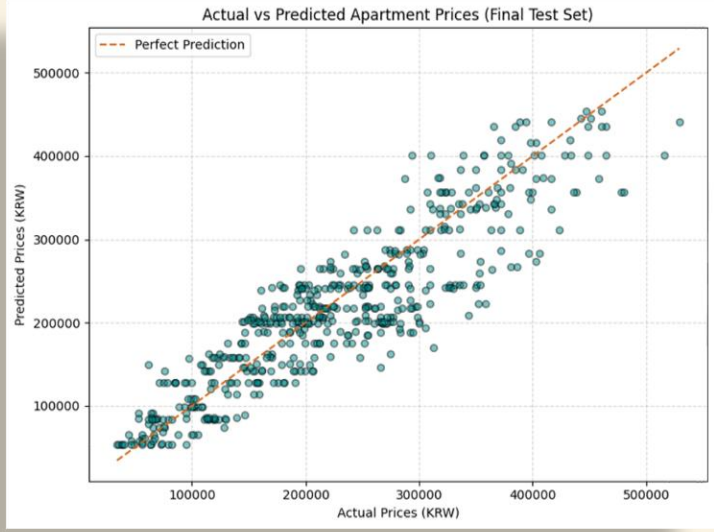
XGBoost uses parallel processing, efficient split finding, and handles sparse data well, enabling fast training on large and complex datasets.

Model Evaluation







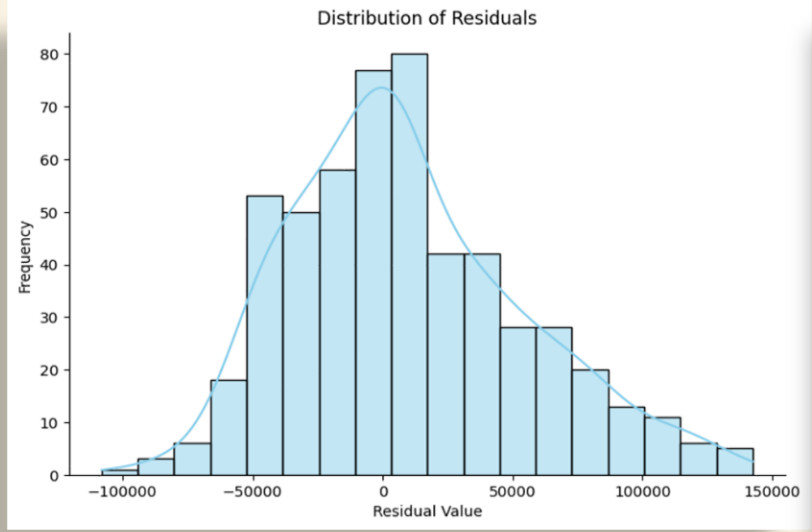


XGBOOST PERFORMANCE METRICS BY MODEL STAGE

- **Real-World Accuracy**
  MAE: 35,101 KRW, MAPE: 17.5% → Predictions are typically within ±35K KRW or ±18%, making results practical and interpretable.
- **Robust & Ready**
  Lower MAE/MAPE with minimal overfitting → CV vs. test consistency confirms strong generalization and deployment readiness.
- **Actual vs. Predicted (Final Test Set)**
  Predictions align closely with actual prices (50–550K KRW range), and $R^2 = 0.80$ confirms the model explains 80% of price variance effectively.
- **Mid-Market Strength**
  Highest accuracy in 150–300K KRW range, covering most transactions and ensuring dependable performance for typical users.

Residual Analysis

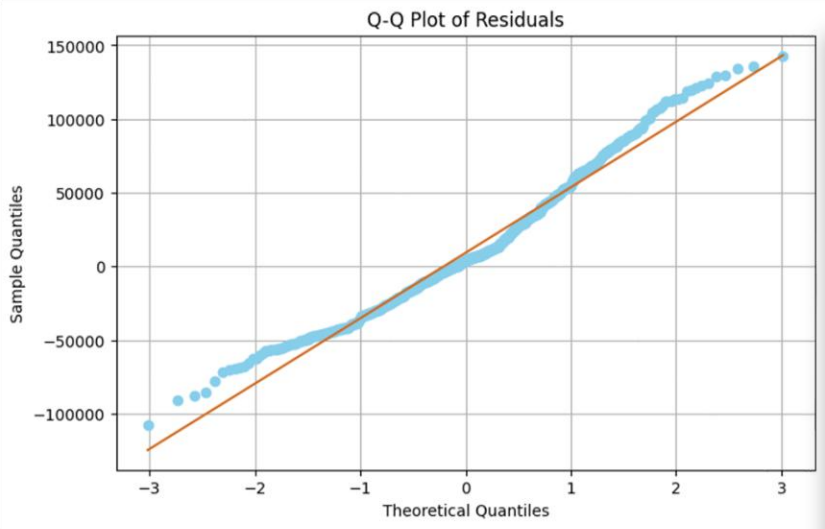
DISTRIBUTION OF RESIDUALS



Accurate Core with Skewed Tails

Residuals center around zero but show a right-skewed tail, indicating underestimation of high-priced units and inflating error metrics.

Q-Q PLOT OF RESIDUALS



Non-Normal Tails in Q–Q Plot

Heavy tails diverge from normality, meaning prediction intervals underestimate uncertainty, especially for luxury properties.

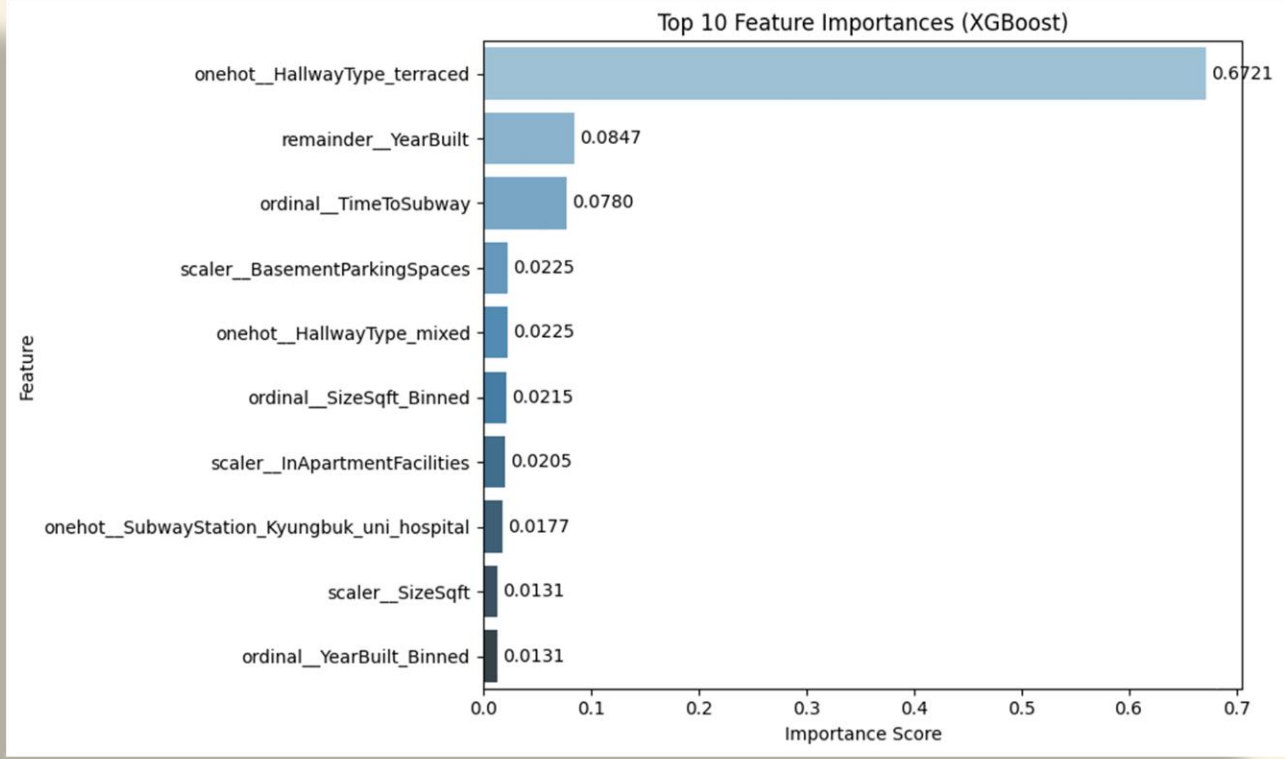
RESIDUALS VS. PREDICTED SALE PRICE

Heteroscedasticity with Price

Residual spread widens with predicted price, confirming increased variance and reduced reliability in high-end segments.

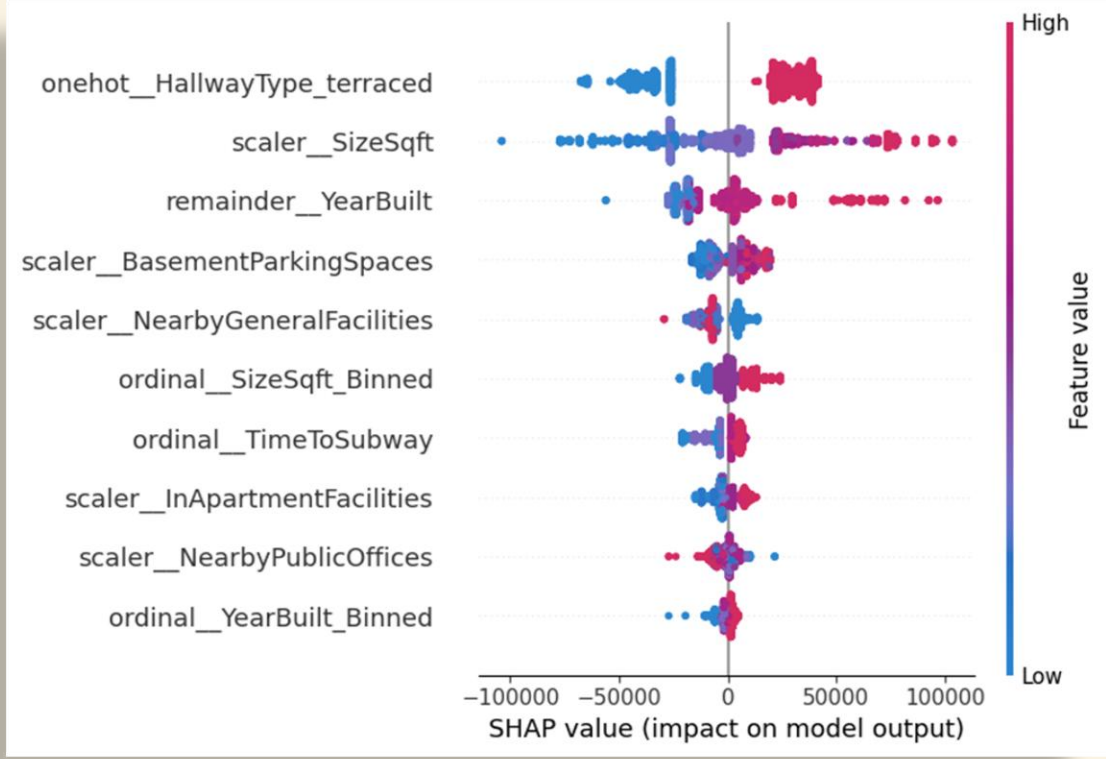
Feature Importance Analysis

GAIN-BASED



‘HallwayType_terraced’, ‘YearBuilt’, and ‘**TimeToSubway’** are **the top gain-based** features driving overall model performance across the dataset.

SHAP



‘HallwayType_terraced’, ‘SizeSqft’, and **‘YearBuilt’** are the **top contributors of apartment price predictions** in Daegu, with the strongest impact on **individual SHAP values.**

Cost-Benefit Analysis

1. ERROR METRICS COMPARISON: WITHOUT ML VS. WITH ML

Metric	Manual Baseline	ML Model	Improvement
MAE	₩66,159,137	₩35,100,890	–₩31,058,247 (–47%)
RMSE	₩71,221,545	₩45,696,090	–₩25,525,455 (–36%)
MAPE	67.58%	17.47%	–50.11 percentage points

2. ESTIMATED MARKET-WIDE FINANCIAL IMPACT

Benefit	Per Unit	Annual Total
MAE reduction (₩31,058,247 × 5,000 units)	₩31,058,247	₩155,291,235,000
RMSE reduction (₩25,525,455 × 5,000 units)	₩25,525,455	₩127,627,275,000

NOTES:

- **Unit Scale Assumption:**
  Price values likely represent **thousands of won**, not individual won — all analyses follow this assumption.
- **Manual Pricing Reference:**
  Estimates use apartment size and **₩726,200/sqm** (Daegu average) to baseline prediction accuracy (without ML).
- **Scaled Error Metrics:**
  MAE, RMSE, and MAPE are scaled to **actual won**, ensuring results reflect true financial impact.
- **Target Market Assumption:**
  From ~42,400 annual housing transactions in Daegu, a conservative estimate of 5,000 apartment listings/year is used, reflecting early adoption within a 17,000 – 25,000 unit market.

3. COST-BENEFIT SUMMARY PER UNIT

Aspect	Cost per Unit	Benefit per Unit	Notes
Under-Pricing Opportunity Loss	₩35,100,000	—	Missed revenue when a property is priced too low
Over-Pricing Holding Cost	—	₩35,100,000	Savings from faster sales and reduced holding or discounting costs
Model Implementation Cost	Development and maintenance	Long-term ROI	Costs spread out over many units as adoption and volume increase

Cost-Benefit Analysis

4. RETURN ON INVESTMENT (ROI) CALCULATIONS

MODEL IMPLEMENTATION COST

Cost Category	Estimate (₩)	Key Explanation
Development & Engineering	₩20,000,000 (one-time)	1–2 months of ML development, data pipeline setup, automation, and testing.
Deployment & Maintenance	₩5,000,000 / year	Covers cloud hosting, APIs, monitoring, and minor updates.
User Training & Support	₩3,000,000 / year	Staff training, user onboarding, and documentation.
Total First-Year Cost	₩28,000,000	₩20M setup + ₩8M for first-year operations.
Ongoing Annual Cost	₩8,000,000 / year	Recurring cloud, support, and maintenance costs.

$$\text{Return on Investment (ROI)} = \frac{\text{Net Return}}{\text{Cost of Investment}}$$

First-Year ROI (including setup cost)

Metric Basis	Annual Benefit	First-Year Cost	Net Benefit (1st Year)	ROI (1st Year)
MAE-based	₩155,300,000,000	₩28,000,000	₩155,272,000,000	5,545×
RMSE-based	₩127,600,000,000	₩28,000,000	₩127,572,000,000	4,556×

Ongoing ROI (excluding setup cost)

Metric Basis	Annual Benefit	Annual Cost	Net Benefit	ROI
MAE-based	₩155,300,000,000	₩8,000,000	₩155,292,000,000	19,411×
RMSE-based	₩127,600,000,000	₩8,000,000	₩127,592,000,000	15,949×

INSIGHTS:

- **First-Year ROI:** Despite ₩28M in setup costs, ROI exceeds 4,000×–5,000×, proving rapid value recovery and strong financial impact.
- **Ongoing ROI:** With only ₩8M in annual costs, ROI reaches over 15,000×–19,000×, showing high long-term efficiency and sustainability.
- **Conclusion:** The model offers both immediate benefits and exceptional long-term economic value for Daegu's apartment market.

CONCLUSION AND RECOMMENDATIONS

The conclusion summarizes key findings and project limitations, while the recommendations focus on pricing strategies and model improvements.

Conclusion

BEST MODEL & PERFORMANCE

1. XGBoost selected after thorough tuning and validation, delivering strong accuracy: MAE ₩35.1M, MAPE 17.5%, RMSE ₩45.7M, $R^2 = 0.80$.
2. Outperforms manual pricing by reducing MAE by 47%, RMSE by 36%, and MAPE by 50%.

BUSINESS IMPACT

1. By improving pricing accuracy in 5,000+ transactions annually, the model could reduce losses by up to ₩155 billion.
2. Reduces manual appraisal effort, accelerates sales, and improves pricing competitiveness.

INTERPRETABILITY & SCALABILITY

1. SHAP Analysis Highlights Key Price Drivers: Terraced hallway, apartment size, and year built have the strongest impact on individual predictions.
2. Feature Importance (Gain-Based): Top features overall are terraced hallway type, year built, and proximity to subway.

STRONG ECONOMIC VALUE

1. Development cost ₩28M with first-year benefit ₩155.3B → over 5,500× ROI.
2. Ongoing annual ROI exceeds 19,000× due to low maintenance and scalable operations.

Limitations

LIMITED SCOPE (REGION & TIME)

Model trained only on Daegu data (1978–2015); may underperform in other regions or under changing market conditions without retraining.

MISSING QUALITATIVE FACTORS

Factors like school quality, buyer sentiment, renovations, and zoning changes weren't included, limiting contextual insight and precision.

NARROW PRICE RANGE

- Predictions are most reliable within the ₩32M–₩586M range.
- Accuracy may drop outside this due to limited data coverage.

MAINTENANCE & UNIT SCALE CAUTION

- Requires ongoing drift detection and retraining to stay accurate.
- Values may be in thousands of won, so unit scale should be clearly understood to avoid misinterpretation.

Recommendations

EXPAND COVERAGE AND ADAPTABILITY

1. Add transaction data from other cities or nationwide.
2. Use transfer learning/domain adaptation to handle new regions and market changes.
3. Monitor recent data to detect and address temporal drift.

ENRICH FEATURE SET

1. Incorporate external data: school ratings, buyer sentiment, renovations, zoning, policy updates.
2. Utilize GIS, census, and local government datasets for deeper insights and explainability.

AUTOMATE MONITORING AND MODEL UPDATES

1. Create alerts if the model's accuracy drops.
2. Schedule periodic retraining with latest data and infrastructure changes.
3. Build a system that automatically handles data updates, retraining, and model deployment.

ENSURE DATA CONSISTENCY AND DOCUMENTATION

1. Make sure all prices use the same unit (like thousands of won).
2. Add automatic checks to avoid mistakes or confusion in the data.

THANK YOU

PROJECT LINKS

Github: Daegu Apartment Price Prediction And Optimization
Tableau Dashboard: Daegu Apartment Price Dashboard